

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (Biotechnology) (Credit System) Degree Examinations**

May - June 2023

20BT601 - PRINCIPLES OF FERMENTER DESIGN AND CONTROL

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I**Marks BT* CO* PO***

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|-------|--|----|-----|---|---|
| 1. a) | Yeast is cultivated in industrial scale fermenter in continuous mode. The quantity handled is 1m^3 per day. The maximum cell growth was observed at optimum temperature and substrate concentration. Suggest necessary supporting component and material of construction required to handle this process. | 10 | L*2 | 1 | 3 |
| b) | The vessel and pipeline network connected to a bioreactor need to be cleaned before charging the fresh batch. In detail explain the cleaning protocol for this system if process involves protein and sugar components at larger quantities. | 10 | L3 | 2 | 3 |
| 2. a) | List and explain the working of any 2 flow valves for handling aseptic condition inside the bioreactor. Draw the diagram and label the parts. | 10 | L2 | 1 | 3 |
| b) | List and explain the prerequisites in the design of a vessel for steam sterilization. | 10 | L3 | 2 | 3 |
| 3. a) | Explain the role of different fermenter components in maintaining the aseptic condition inside the bioreactor. | 10 | L2 | 1 | 3 |
| b) | In detail explain the survival curve and explain the significance of D and Z value in heat sterilization of bioreactor. | 10 | L2 | 2 | 3 |

Unit – II

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|-------|--|----|----|---|---|
| 4. a) | In a continuous mode submerged liquid fermenter (SLF) cell concentration to be maintained at optimum at any given time. Suggest a suitable reactor modification and derive an expression to calculate the cell concentration. | 10 | L3 | 3 | 2 |
| b) | A bioprocess in which oxygen requirement for aerobic cell culture is maintained by continuous sparging of sterile air. Derive an expression to determine the rate of mass transfer between liquid culture and gaseous oxygen. | 10 | L3 | 4 | 2 |
| 5. a) | Immobilized lactase is used to hydrolyze lactose in dairy waste to glucose and galactose. Enzyme is immobilized in resin particles and packed into a 0.5m^3 column. The total effectiveness factor for the system is close to unity; K_m for the immobilized enzyme is 1.32kg/m^3 , V_{max} is $45\text{kg/m}^3\cdot\text{h}$. The lactose concentration in the feed stream is 9.5kg/m^3 ; a substrate conversion of 98% is required. The column is operated with plug flow for a total of 310 days per year.
i) At what flow rate should the reactor be operated?
ii) How many tons of glucose are produced per year?
Data: Molecular weight: Lactose=342, Glucose=180 | 10 | L3 | 3 | 2 |
| b) | In detail k_{La} measurement techniques in bioreactor. | 10 | L2 | 4 | 3 |

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|-------|---|----|----|---|---|
| 6. a) | Microbial cultivation is carried out in fed batch reactor having volume V. Derive an expression to determine cell concentration in reactor. | 10 | L3 | 3 | 2 |
| b) | List and explain the factors affecting the oxygen solubility in fermenter. | 10 | L2 | 4 | 3 |

Unit – III

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|-------|--|----|----|---|---|
| 7. a) | With neat, labeled diagram explain the construction and working of calomel electrode. | 10 | L2 | 5 | 2 |
| b) | Explain the response of proportional Derivative (PD) and proportional Integral (PI) controller to a linear input. Also discuss the effect of controller setting on controller response. | 10 | L3 | 5 | 2 |
| 8. a) | Explain the construction and working of polarographic sensor in measuring the dissolved oxygen in bioreactor. | 10 | L2 | 5 | 3 |
| b) | A proportional derivative (PI) controller is used to control first order system with first order measuring element. Determine the expression for the offset for servo-mechanism control problem and discuss the effect of integral time (τ_i) on controller response. | 10 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
